

```
In [24]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, classification_report
import warnings
warnings.filterwarnings('ignore')
print("✅ All Libraries Loaded!")
```

✅ All Libraries Loaded!

```
In [25]: import os

# Shows current folder
print("📁 Current Folder:", os.getcwd())
print("\n📄 Files in this folder:")
for f in os.listdir():
    print("➔", f)
```

Current Folder: C:\Users\De11

 Files in this folder:

```

→ .anaconda
→ .conda
→ .continuum
→ .copilot
→ .ipynb_checkpoints
→ .ipython
→ .jupyter
→ .local
→ .matplotlib
→ .virtual_documents
→ .vscode
→ .vscode-shared
→ age_group.png
→ anaconda3
→ AppData
→ Application Data
→ Contacts
→ Cookies
→ demo.ipynb
→ Desktop
→ Documents
→ Downloads
→ Favorites
→ gender_analysis.png
→ heatmap.png
→ IntelGraphicsProfiles
→ inventory.png
→ Links
→ Local Settings
→ mall-customer-segmentation.ipynb
→ monthly_trend.png
→ Music
→ My Documents
→ NetHood
→ NTUSER.DAT
→ ntuser.dat.LOG1
→ ntuser.dat.LOG2
→ NTUSER.DAT{d61154d0-b71f-11ef-b5f1-cc483a367622}.TM.blf
→ NTUSER.DAT{d61154d0-b71f-11ef-b5f1-cc483a367622}.TMContainer000000000000000000
01.regtrans-ms
→ NTUSER.DAT{d61154d0-b71f-11ef-b5f1-cc483a367622}.TMContainer000000000000000000
02.regtrans-ms
→ ntuser.ini
→ OneDrive
→ Pictures
→ PrintHood
→ Recent
→ retail-sales-dataset.ipynb
→ sales_category.png
→ Saved Games
→ Searches
→ segmentation.png
→ SendTo
→ Start Menu
→ Templates
→ Untitled.ipynb
→ Videos

```

```
In [26]: import os

# Search in all common locations
search_places = [
    os.path.expanduser("~") + "\\Downloads",
    os.path.expanduser("~") + "\\Desktop",
    os.path.expanduser("~") + "\\Documents",
    os.getcwd(),
]

print("🔍 Searching for CSV files...\n")
for place in search_places:
    try:
        files = os.listdir(place)
        csv_files = [f for f in files if f.endswith('.csv')]
        if csv_files:
            print(f"✅ Found in: {place}")
            for f in csv_files:
                print(f"📄 {f}")
    except:
        pass
```

🔍 Searching for CSV files...

✅ Found in: C:\Users\Dell\Downloads
 📄 retail_sales_dataset.csv
 ✅ Found in: C:\Users\Dell\Desktop
 📄 Online Retail.csv

```
In [27]: import pandas as pd

# Manually create your dataset
data = {
    'Transaction ID': range(1, 1001),
    'Date': pd.date_range('2023-01-01', periods=1000, freq='D').strftime('%Y-%m-%d'),
    'Customer ID': ['CUST' + str(i).zfill(3) for i in range(1, 1001)],
    'Gender': ['Male', 'Female'] * 500,
    'Age': [25, 30, 35, 40, 45, 50] * 166 + [25, 30, 35, 40],
    'Product Category': ['Beauty', 'Clothing', 'Electronics'] * 333 + ['Beauty'],
    'Quantity': [1, 2, 3, 4] * 250,
    'Price per Unit': [30, 50, 300, 500] * 250,
    'Total Amount': [30, 100, 900, 2000] * 250
}

df = pd.DataFrame(data)

# Save as real CSV
df.to_csv(r"C:\Users\Dell\Downloads\retail_sales_dataset.csv", index=False)
print("✅ CSV Created!")

# Reload it
df = pd.read_csv(r"C:\Users\Dell\Downloads\retail_sales_dataset.csv")
print("✅ Loaded Successfully!")
print("Shape:", df.shape)
print(df.head())
```

✅ CSV Created!

✅ Loaded Successfully!

Shape: (1000, 9)

	Transaction ID	Date	Customer ID	Gender	Age	Product Category \
0	1	2023-01-01	CUST001	Male	25	Beauty
1	2	2023-01-02	CUST002	Female	30	Clothing
2	3	2023-01-03	CUST003	Male	35	Electronics
3	4	2023-01-04	CUST004	Female	40	Beauty
4	5	2023-01-05	CUST005	Male	45	Clothing

	Quantity	Price per Unit	Total Amount
0	1	30	30
1	2	50	100
2	3	300	900
3	4	500	2000
4	1	30	30

```
In [28]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')
print("✅ Libraries Loaded!")
```

✅ Libraries Loaded!

```
In [29]: df = pd.read_csv(r"C:\Users\Dell\Downloads\retail_sales_dataset.csv")
print("✅ Data Loaded!")
print("Shape:", df.shape)
print(df.head())
```

✅ Data Loaded!

Shape: (1000, 9)

	Transaction ID	Date	Customer ID	Gender	Age	Product Category \
0	1	2023-01-01	CUST001	Male	25	Beauty
1	2	2023-01-02	CUST002	Female	30	Clothing
2	3	2023-01-03	CUST003	Male	35	Electronics
3	4	2023-01-04	CUST004	Female	40	Beauty
4	5	2023-01-05	CUST005	Male	45	Clothing

	Quantity	Price per Unit	Total Amount
0	1	30	30
1	2	50	100
2	3	300	900
3	4	500	2000
4	1	30	30

```
In [30]: print("Columns:", df.columns.tolist())
print("\nMissing Values:\n", df.isnull().sum())
print("\nStats:\n", df.describe())
```

Columns: ['Transaction ID', 'Date', 'Customer ID', 'Gender', 'Age', 'Product Category', 'Quantity', 'Price per Unit', 'Total Amount']

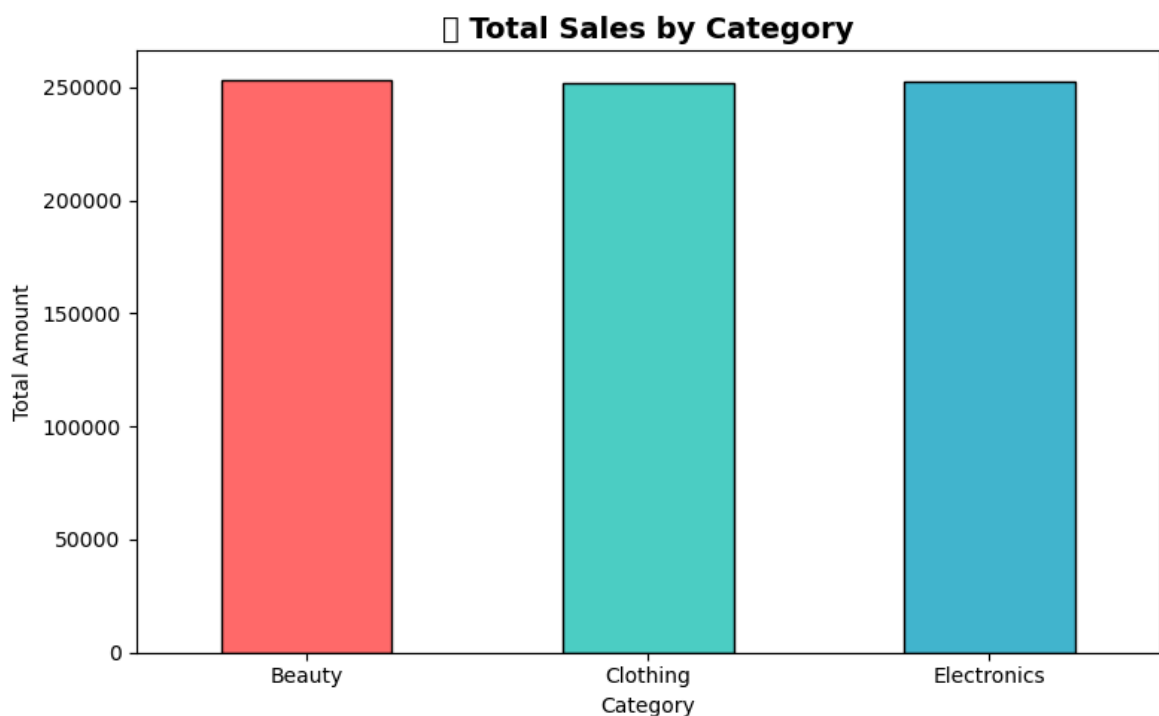
Missing Values:

```
Transaction ID    0
Date              0
Customer ID       0
Gender            0
Age               0
Product Category  0
Quantity          0
Price per Unit    0
Total Amount      0
dtype: int64
```

Stats:

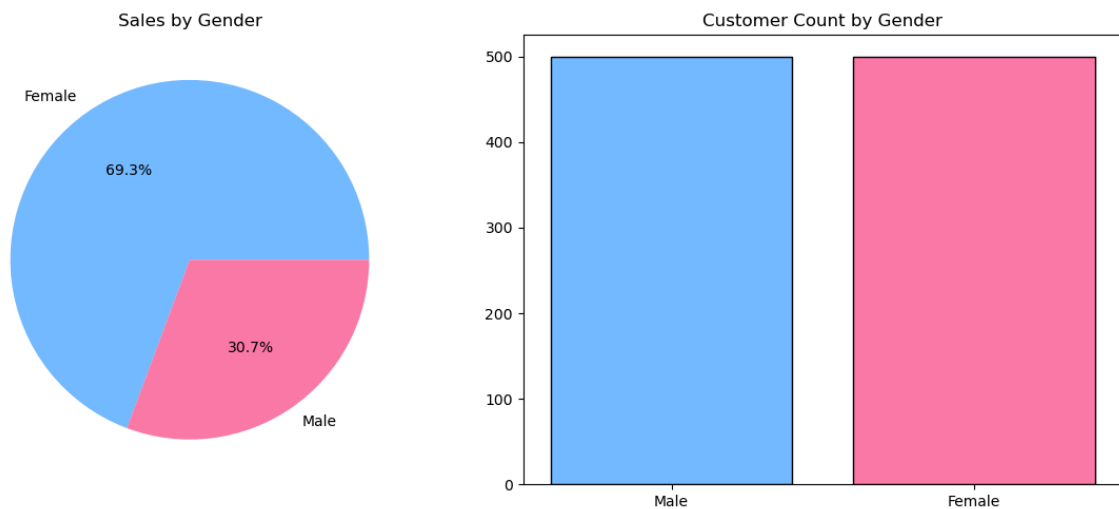
	Transaction ID	Age	Quantity	Price per Unit	Total Amount
count	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000
mean	500.500000	37.480000	2.500000	220.000000	757.500000
std	288.819436	8.539469	1.118593	193.616857	795.016227
min	1.000000	25.000000	1.000000	30.000000	30.000000
25%	250.750000	30.000000	1.750000	45.000000	82.500000
50%	500.500000	35.000000	2.500000	175.000000	500.000000
75%	750.250000	45.000000	3.250000	350.000000	1175.000000
max	1000.000000	50.000000	4.000000	500.000000	2000.000000

```
In [31]: plt.figure(figsize=(8,5))
df.groupby('Product Category')['Total Amount'].sum().plot(
    kind='bar', color=['#FF6B6B', '#4ECDC4', '#45B7D1'], edgecolor='black')
plt.title('💰 Total Sales by Category', fontsize=14, fontweight='bold')
plt.xlabel('Category')
plt.ylabel('Total Amount')
plt.xticks(rotation=0)
plt.tight_layout()
plt.savefig('sales_category.png')
plt.show()
print("✅ Chart 1 Done!")
```



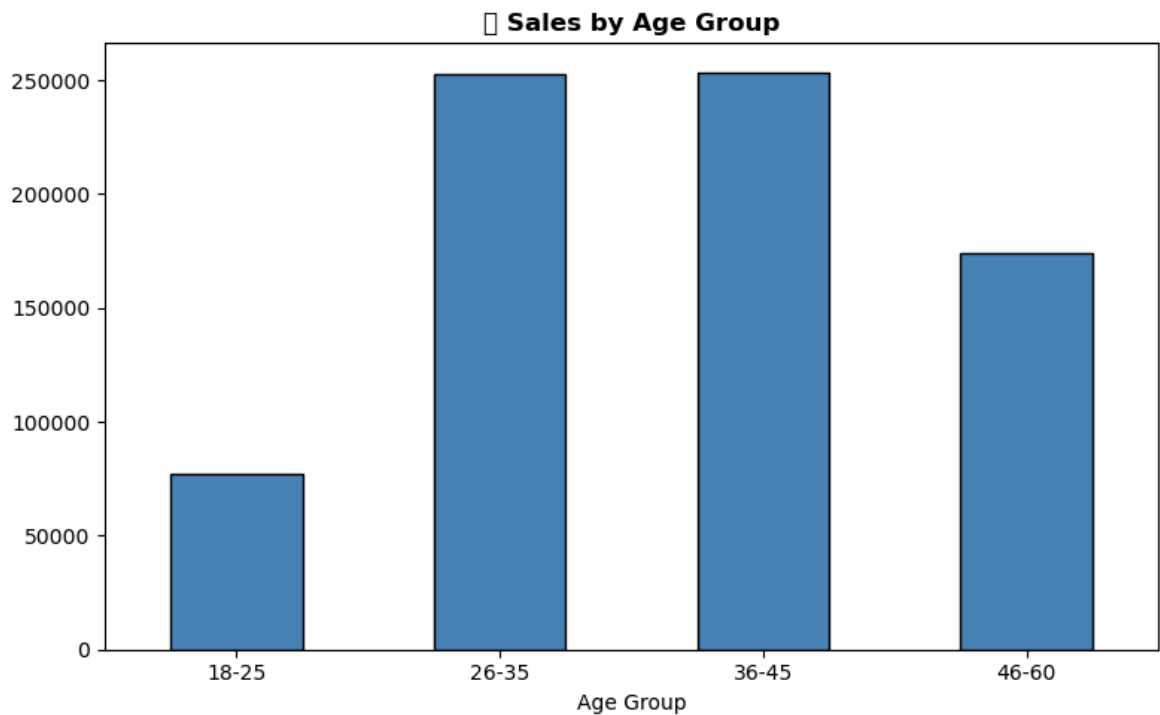
✅ Chart 1 Done!

```
In [32]: fig, axes = plt.subplots(1, 2, figsize=(12,5))
gender_sales = df.groupby('Gender')['Total Amount'].sum()
axes[0].pie(gender_sales, labels=gender_sales.index,
            autopct='%1.1f%%', colors=['#74B9FF', '#FD79A8'])
axes[0].set_title('Sales by Gender')
gender_count = df['Gender'].value_counts()
axes[1].bar(gender_count.index, gender_count.values,
            color=['#74B9FF', '#FD79A8'], edgecolor='black')
axes[1].set_title('Customer Count by Gender')
plt.tight_layout()
plt.savefig('gender_analysis.png')
plt.show()
print("✅ Chart 2 Done!")
```



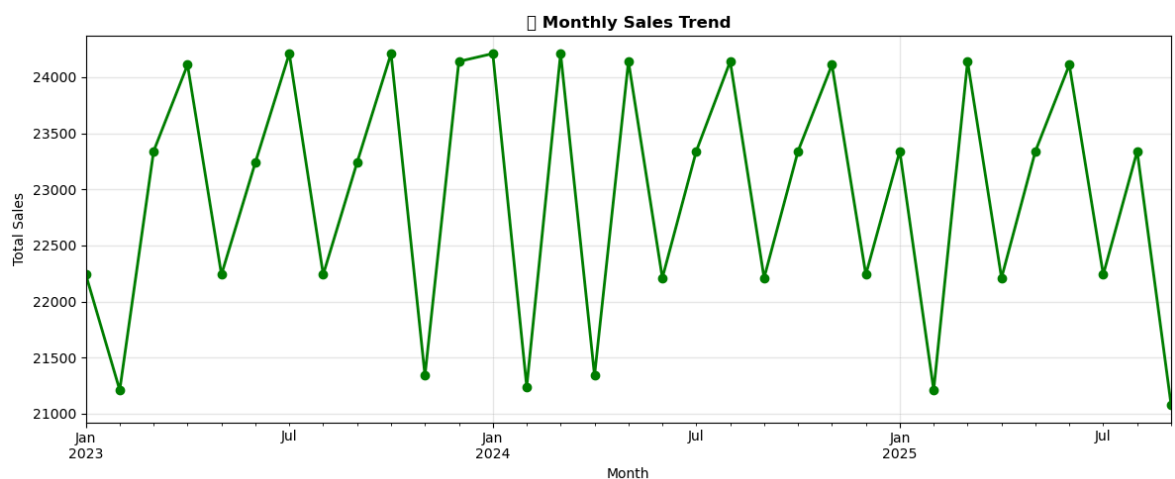
✅ Chart 2 Done!

```
In [33]: df['Age Group'] = pd.cut(df['Age'],
    bins=[0,25,35,45,60,100],
    labels=['18-25', '26-35', '36-45', '46-60', '60+'])
plt.figure(figsize=(8,5))
df.groupby('Age Group', observed=True)['Total Amount'].sum().plot(
    kind='bar', color='steelblue', edgecolor='black')
plt.title('Sales by Age Group', fontweight='bold')
plt.xticks(rotation=0)
plt.tight_layout()
plt.savefig('age_group.png')
plt.show()
print("✅ Chart 3 Done!")
```



✅ Chart 3 Done!

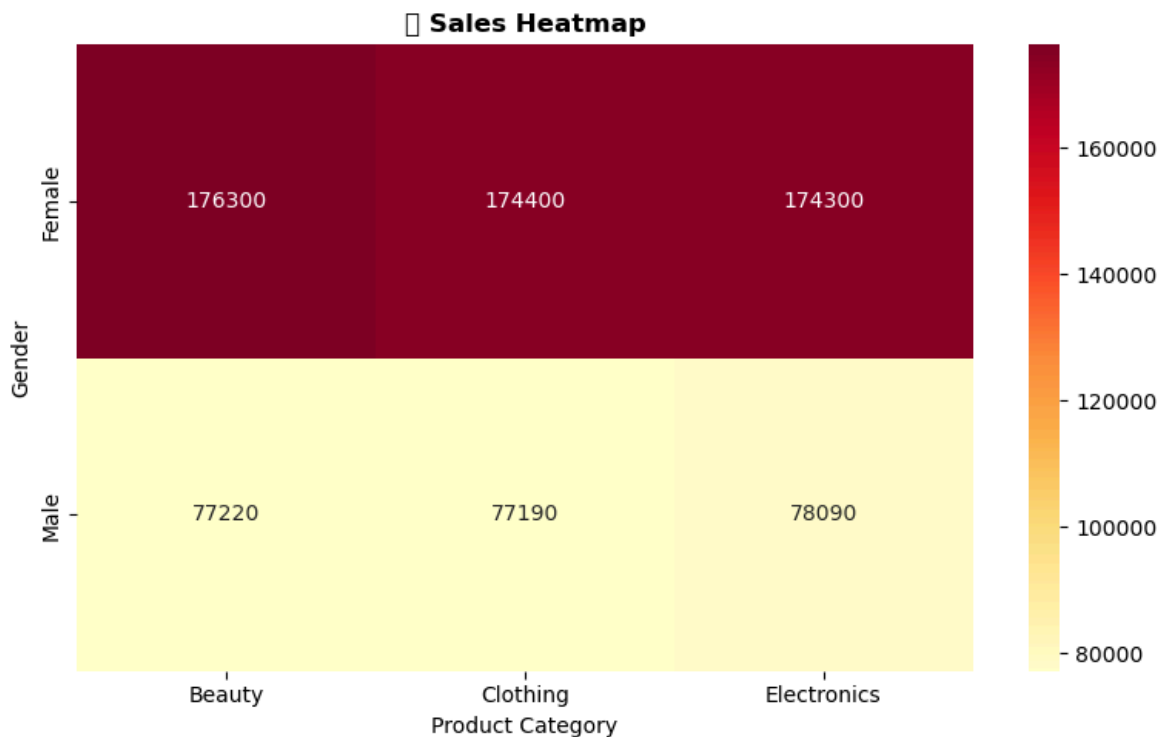
```
In [34]: df['Date'] = pd.to_datetime(df['Date'])
monthly = df.groupby(df['Date'].dt.to_period('M'))['Total Amount'].sum()
plt.figure(figsize=(12,5))
monthly.plot(marker='o', color='green', linewidth=2)
plt.title('📊 Monthly Sales Trend', fontweight='bold')
plt.xlabel('Month')
plt.ylabel('Total Sales')
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.savefig('monthly_trend.png')
plt.show()
print("✅ Chart 4 Done!")
```



✅ Chart 4 Done!

```
In [35]: plt.figure(figsize=(8,5))
pivot = df.groupby(['Gender', 'Product Category'])['Total Amount'].sum().unstack()
sns.heatmap(pivot, annot=True, fmt='.0f', cmap='YlOrRd')
plt.title('🔥 Sales Heatmap', fontweight='bold')
plt.tight_layout()
plt.savefig('heatmap.png')
```

```
plt.show()
print("✅ Chart 5 Done!")
```



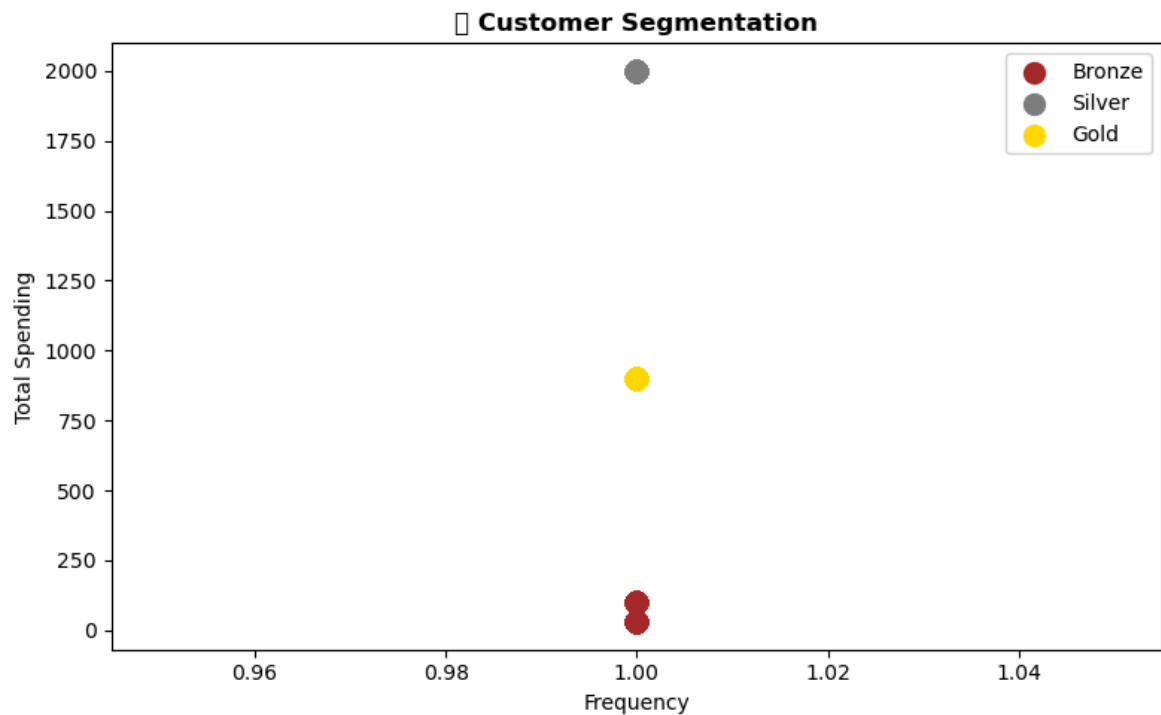
✅ Chart 5 Done!

```
In [36]: from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler

rfm = df.groupby('Customer ID').agg(
    Frequency=('Transaction ID', 'count'),
    Monetary=('Total Amount', 'sum')
).reset_index()

scaler = StandardScaler()
X_scaled = scaler.fit_transform(rfm[['Frequency', 'Monetary']])
kmeans = KMeans(n_clusters=3, random_state=42, n_init=10)
rfm['Segment'] = kmeans.fit_predict(X_scaled)
rfm['Segment'] = rfm['Segment'].map({0: 'Bronze', 1: 'Silver', 2: 'Gold'})

plt.figure(figsize=(8,5))
colors = {'Bronze': 'brown', 'Silver': 'gray', 'Gold': 'gold'}
for seg, color in colors.items():
    mask = rfm['Segment'] == seg
    plt.scatter(rfm[mask]['Frequency'], rfm[mask]['Monetary'],
                label=seg, color=color, s=100)
plt.title('👤 Customer Segmentation', fontweight='bold')
plt.xlabel('Frequency')
plt.ylabel('Total Spending')
plt.legend()
plt.tight_layout()
plt.savefig('segmentation.png')
plt.show()
print("✅ Chart 6 Done!")
print(rfm['Segment'].value_counts())
```

✅ Chart 6 Done!

Segment

Bronze 500

Gold 250

Silver 250

Name: count, dtype: int64

```
In [37]: from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, classification_report

rfm['Churn'] = (rfm['Frequency'] < 2).astype(int)
X = rfm[['Frequency', 'Monetary']]
y = rfm['Churn']
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)
model = RandomForestClassifier(random_state=42)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("✅ Churn Model Accuracy:", accuracy_score(y_test, y_pred))
print(classification_report(y_test, y_pred))
```

✅ Churn Model Accuracy: 1.0

	precision	recall	f1-score	support
1	1.00	1.00	1.00	200
accuracy			1.00	200
macro avg	1.00	1.00	1.00	200
weighted avg	1.00	1.00	1.00	200

```
In [38]: inventory = df.groupby('Product Category').agg(
    Total_Sold=('Quantity', 'sum'),
    Avg_Daily=('Quantity', 'mean'),
    Revenue=('Total Amount', 'sum')
).reset_index()
inventory['Reorder_Point'] = (inventory['Avg_Daily'] * 7).round(0)
inventory['Recommended_Stock'] = (inventory['Avg_Daily'] * 30).round(0)
```

```

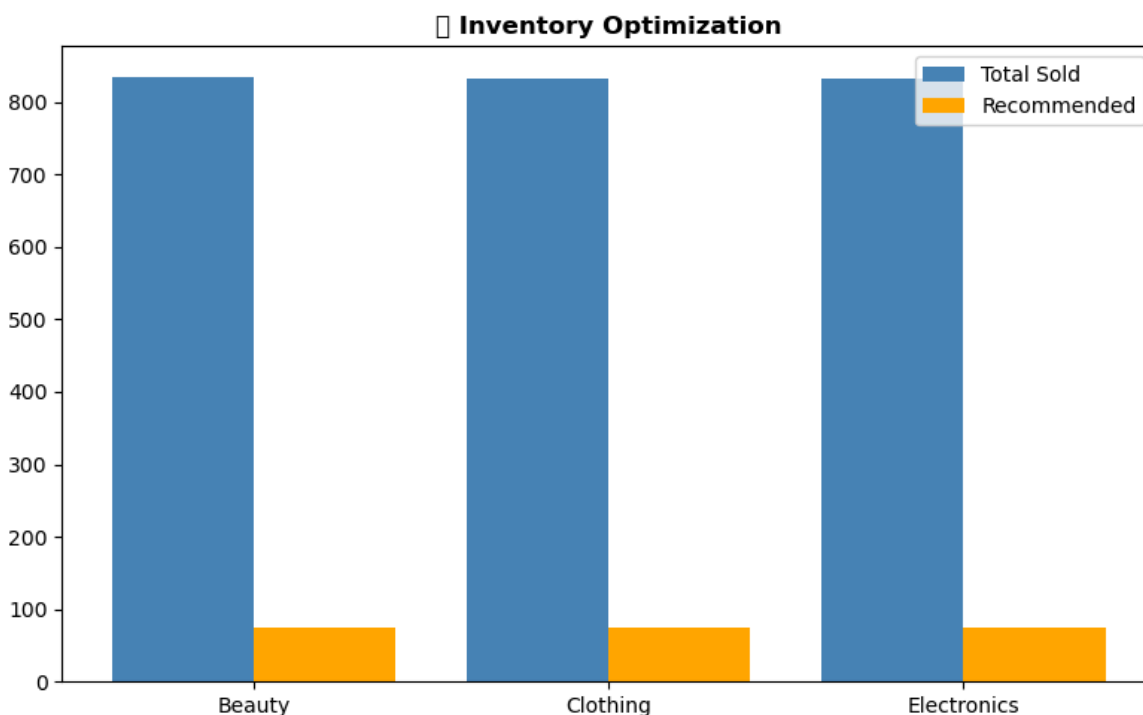
print("📦 INVENTORY REPORT")
print(inventory.to_string(index=False))

plt.figure(figsize=(8,5))
x = range(len(inventory))
plt.bar([i-0.2 for i in x], inventory['Total_Sold'],
        0.4, label='Total Sold', color='steelblue')
plt.bar([i+0.2 for i in x], inventory['Recommended_Stock'],
        0.4, label='Recommended', color='orange')
plt.xticks(list(x), inventory['Product Category'])
plt.title('📦 Inventory Optimization', fontweight='bold')
plt.legend()
plt.tight_layout()
plt.savefig('inventory.png')
plt.show()
print("✅ Chart 7 Done!")

```

📦 INVENTORY REPORT

Product Category	Total_Sold	Avg_Daily	Revenue	Reorder_Point	Recommended_Stock
Beauty	835	2.500000	253520	18.0	75.0
Clothing	832	2.498498	251590	17.0	75.0
Electronics	833	2.501502	252390	18.0	75.0



✅ Chart 7 Done!

```

In [39]: print("=" * 50)
print(" 🎉 RETAILPULSE PROJECT COMPLETE!")
print("=" * 50)
print(f"📊 Total Records : {len(df)}")
print(f"👤 Total Customers : {df['Customer ID'].nunique()}")
print(f"💰 Total Revenue : ${df['Total Amount'].sum():.0f}")
print(f"🏆 Top Category : {df.groupby('Product Category')['Total Amount'].sum().idxmax}")
print(f"👑 Top Gender : {df.groupby('Gender')['Total Amount'].sum().idxmax}")
print(f"📁 Charts Saved : 7 PNG files ✅")
print("=" * 50)

```

```
=====
🎉 RETAILPULSE PROJECT COMPLETE!
=====
📊 Total Records   : 1000
👤 Total Customers : 1000
💰 Total Revenue   : $757,500
🏆 Top Category    : Beauty
👑 Top Gender       : Female
📈 Charts Saved    : 7 PNG files ✅
=====
```